Q1 - 26 July - Shift 1

A screw gauge of pitch 0.5mm is used to measure the diameter of uniform wire of length 6.8cm, the main scale reading is 1.5 mm and circular scale reading is 7. The calculated curved surface area of wire to appropriate significant figures is :

[Screw gauge has 50 divisions on the circular scale]

(A) $6.8 cm^2$ (B) $3.4 cm^2$

(C) $3.9 cm^2$ (D) $2.4 cm^2$

Space for your notes:

Q2 - 26 July - Shift 2

In a Vernier Calipers. 10 divisions of Vernier scale is equal to the 9 divisions of main scale. When both jaws of Vernier calipers touch each other, the zero of the Vernier scale is shifted to the left of zero of the main scale and 4^{th} Vernier scale division exactly coincides with the main scale reading. One main scale division is equal to 1 mm. While measuring diameter of a spherical body, the body is held between two jaws. It is now observed that zero of the Vernier scale lies between 30 and 31 divisions of main scale reading and 6^{th} Vernier scale division exactly. coincides with the main scale reading. The diameter of the spherical body will be :

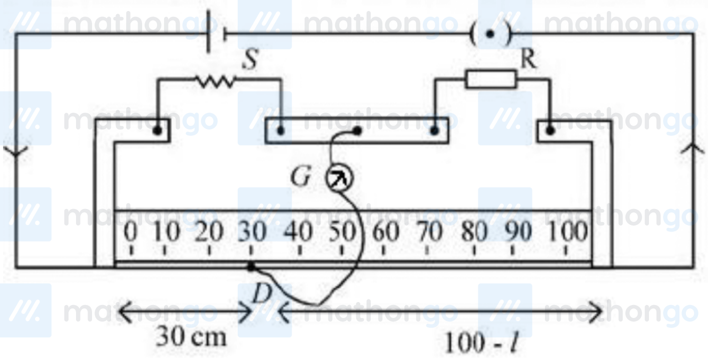
(A) 3.02 cm (B) 3.06 cm

(C) 3.10 cm (D) 3.20 cm

Space for your notes:

Q3 - 27 July - Shift 1

In meter bridge experiment for measuring unknown resistance 'S', the null point is obtained at a distance 30 cm from the left side as shown at point D. If R is 5.6 kΩ, then the value of unknown resistance 'S' will be ________ Ω.



Space for your notes:

Q4 - 28 July - Shift 2

In an experiment with a convex lens. The plot of the image distance (v') against the object distance (μ') measured from the focus gives a curve v' μ' = 225. If all the distances are measured in cm. The magnitude of the focal length of the lens is ……….. cm.

Space for your notes:

Q5 - 29 July - Shift 1

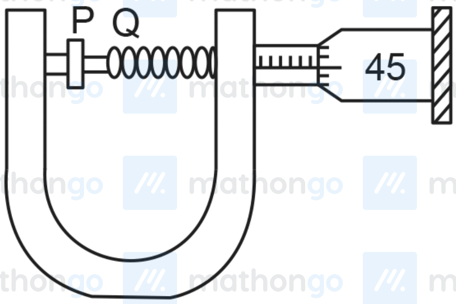
A travelling microscope has 20 divisions per cm on the main scale while its Vernier scale has total 50 divisions and 25 Vernier scale divisions are equal to 24 main scale divisions, what is the least count of the travelling microscope ?

(A) 0.001 cm (B) 0.002 mm
(C) 0.002 cm (D) 0.005 cm

Space for your notes:

Q6 - 29 July - Shift 1

In an experiment to find out the diameter of wire using screw gauge, the following observation were noted :



(a) Screw moves 0.5 mm on main scale in one complete rotation
(b) Total divisions on circular scale = 50
(c) Main scale reading is 2.5 mm
(d) 45^{th} division of circular scale is in the pitch line
(e) Instrument has 0.03 mm negative error

Then the diameter of wire is :

(A) 2.92 mm (B) 2.54 mm
(C) 2.98 mm (D) 3.45 mm

Space for your notes:

Answer Key

Q1 (B) **Q2** (C) **Q3** (2400) **Q4** (15)

Q5 (C) **Q6** (C)

Q1 (B)

$$L.C. = \frac{P}{N} = \frac{0.5\text{mm}}{50} = 0.01\text{ mm}$$

Length of wire = 6.8 cm

Diameter of wire = 1.5 mm + 7 × L.C

= 1.5 mm + 7 × .01 = 1.57 mm

Curved surface area = $\pi D\ell$

$= 3.14 \times 6.8 \times 1.57 \times 10^{-1}\text{ cm}^2$

$= 3.352\text{ cm}^2 = 3.4\text{ cm}^2$

Q2 (C)

1 M.S.D = 1mm

9 M.S.D = 10 V.S.D

1 V.S.D = 0.9 M.S.D = 0.9 mm

L.C of vernier caliper = 1–0.9 = 0.1 mm = 0.01 cm

zero error =–(10–4)×0.1mm = – 0.6 mm

Reading = M.S.R + V.S.R – Zero error

= 3cm + 6 × 0.01 – [–0.06]

= 3 + 0.06 + 0.06

= 3.12 cm

Nearest given answer in the options is 3.10

Q3 (2400)

$$\frac{S}{30} = \frac{5.6 \times 10^3}{70}$$

$$S = \frac{3}{7} \times 5.6 \times 10^3 = 2400$$

Q4 (15)

$vu = f^2$ (by Newton's formula)

$f^2 = 225$

$f = 15$ cm

Q5 (C)

$$1\ \text{MSD} = \frac{1}{20}\ \text{cm}$$

$$1\ \text{VSD} = \frac{24}{25}\ \text{MSD} = \frac{24}{25} \times \frac{1}{20}\ \text{cm}$$

$$\therefore \text{Least count} = \frac{1}{20}\left(1 - \frac{24}{25}\right)\text{cm}$$

$$= \frac{1}{20} \times \frac{1}{25} = \frac{1}{500}\ \text{cm}$$

$$= 0.002\ \text{cm}$$

Q6 (C)

$\text{MSR} = 2.5\ \text{mm}$

$\text{CSR} = 45 \times \frac{0.5}{50}\ \text{mm}$

$= 0.45\ \text{mm}$

Diameter reading = MSR + CSR – zero error

$= 2.5 + 0.45 - (-0.03)$

$= 2.98\ \text{mm}$